



Linux Sudoers File – Complete Notes

1 What is Sudo?

- **sudo = Super User Do**
- It allows a normal user to run commands with **admin (root) privileges**

Example:

```
sudo apt update
```

2 What is Sudoers File?

- The **sudoers file** controls who can use sudo
- It defines:
 - Which user can run commands
 - Which commands they can run
 - On which system

File Location:

```
/etc/sudoers
```

3 Why is Sudoers Important?

- Provides security
- Prevents unauthorized access
- Allows controlled admin access

👉 Without sudoers, users cannot run admin commands

4 How to Edit Sudoers File

⚠️ Never edit directly using **vi** or **nano**

Use this command:

`sudo visudo`

- It checks syntax before saving
 - Prevents system errors
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5 Basic Sudoers Syntax

Example:

`username ALL=(ALL) ALL`

Meaning:

Part	Meaning
username	User name
ALL	All hosts
(ALL)	All users
ALL	All commands

👉 This means: user can run all commands as root

6 Common Examples

Full sudo access:

`nishant ALL=(ALL) ALL`

Allow specific command only:

nishant ALL=(ALL) /bin/l*s*

👉 User can only run *ls* with sudo

Passwordless sudo:

nishant ALL=(ALL) NOPASSWD: ALL

👉 No password required

7 Sudo Group

- In Ubuntu, users in the **sudo group** get sudo access

Command:

sudo usermod -aG sudo username

8 Best Practices

- Always use *visudo*
 - Do not give full access to all users
 - Use limited permissions when possible
 - Avoid *NOPASSWD* unless necessary
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9 Real-Life Example

- Normal user = Employee
- Root user = Admin/Manager

👉 Sudoers file = Permission list

👉 It decides:

- Who can do admin work
- What work they can do

10 What is `/etc/sudoers.d`?

- `/etc/sudoers.d` is a directory in Linux
- It stores **extra sudo rule files**
- These files work with the main sudoers file

Paths:

`/etc/sudoers`

`/etc/sudoers.d/`

11 Why Use `/etc/sudoers.d`?

- Avoid editing the main sudoers file
- Keep configuration safe and organized
- Manage permissions for different users separately

👉 If the main file breaks → system problem

👉 This method is safer

12 How It Works

- The main sudoers file automatically includes this directory
- All files inside `/etc/sudoers.d/` are read by the system

👉 You can create separate files for each user or group

13 Example

Create a file:

```
sudo visudo -f /etc/sudoers.d/nishant
```

Add rule:

```
nishant ALL=(ALL) ALL
```

👉 This gives sudo access to user **nishant**

14 Important Rules

- Always use `visudo -f`
- File permission must be **0440**

Set permission:

```
sudo chmod 440 /etc/sudoers.d/nishant
```

👉 Wrong syntax → sudo will stop working ❌

15 Benefits

- Safe configuration
 - Easy to manage
 - No need to edit main file
 - Useful in automation (Ansible, Terraform)
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✅ Final Summary

- `sudo` → run commands as admin
- `/etc/sudoers` → main configuration file
- `/etc/sudoers.d` → extra rules directory
- Use `visudo` or `visudo -f`
- Improves system security

👉 Best practice: Use `/etc/sudoers.d` instead of editing the main file
